

Loong Kuan Lee

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Education

- 2019–2023 **Doctor of Philosophy (PhD)**, *Monash University*, Melbourne
thesis *Computing Divergences between High Dimensional Graphical Models*
supervisors Geoff Webb, Daniel Schmidt, Nico Piatkowski
- 2015–2018 **Bachelor of Informatics and Computation Advanced (Honours)**, *Monash University*, Melbourne, GPA 3.9/4.0, Graduated with First Class Honours.
thesis *Generating Concept Drift by Shuffling Instances.*
majors Computer Science, Probability & Statistics

Experience

- 2023–present **Research Associate**, *Fraunhofer IAIS*, Sankt Augustin, Germany
Researching **quantum optimisation**, both in self-directed research and in applied projects delivered jointly with industry partners.
 - Proposed and developed a framework for incorporating inequality constraints into **QUBO** formulations without needing additional variables or parameter tuning. To appear at **IEEE QCE 2026**.
 - Proposed and studied how the competing objectives of a multi-objective problem can be scaled and combined so that the resulting **QUBO** preserves the intended trade-off. Published at **IEEE QCE 2025**.
 - Formulated power-system redispatch as a multi-objective **QUBO** and evaluated it on **quantum annealing** hardware, in a joint project with a professional-services firm and a German energy utility company. Published at **IEEE DSAA 2025**.
 - Contributed to a hybrid quantum-classical method for multi-agent pathfinding. Published at **ICML 2025**.
- 2016–2017 **Undergraduate Research Assistant**, *Monash University*, Melbourne
Researched how to measure and visualise **concept drift** in both streaming and static data. Built a system to incrementally measure changes to the probability distributions of a data set over time, along with a companion web application.
- 2017 **Quality Assurance Intern**, *Carsales*, Melbourne
Created automated tests for the backend APIs being developed during migration to a microservice architecture. Liaised between multiple teams to ensure sufficient test coverage.
- 2016 **Winter Research Intern**, *Agilent & Monash University*, Melbourne
Developed application to compare and analyse large groups of timeseries data over the same domain based on specifications gathered from stakeholders at Agilent.

Skills

- quantum QUBO formulation, multi-objective optimisation, quantum annealing (D-Wave Ocean), gate-model circuits (Qiskit), classical mixed-integer programming solvers

ML probabilistic graphical models, divergence estimation, concept drift
languages Python, Julia, Rust, C++, JavaScript, Guile Scheme
tools GNU Emacs, Git, LaTeX, Guix, Linux

Featured Publications

Loong Kuan Lee, Harsha Nagarajan, Thore Gerlach, Sascha Mücke, Ragavi Krishnamoorthy, and Nico Piatkowski. Hardware-aware QUBO reformulation of constrained binary optimization via the Walsh-Fourier transform. In *IEEE QCE*, 2026. To appear.

Loong Kuan Lee, Thore Gerlach, and Nico Piatkowski. Standardization of multi-objective QUBOs. In *IEEE QCE*, volume 01, pages 58–64, 2025.

Thore Gerlach, Loong Kuan Lee, Frederic Barbaresco, and Nico Piatkowski. Hybrid quantum-classical multi-agent pathfinding. In *ICML*, volume 267, pages 19161–19171, 2025.

Loong Kuan Lee, Nico Piatkowski, François Petitjean, and Geoffrey I. Webb. Computing divergences between discrete decomposable models. *AAAI*, 37(10):12243–12251, 2023.

Other Publications

Loong Kuan Lee, Ragavi Krishnamoorthy, and Nico Piatkowski. High-order Markov blanket discovery via a k-order relaxation of the faithfulness assumption. In *UAI*, volume 337, pages 3338–3354, 2026.

Sascha Mücke, Raoul Heese, Thore Gerlach, David Biesner, Loong Kuan Lee, and Nico Piatkowski. Quantum adiabatic generation of human-like passwords. In *IEEE QCE*, pages 1516–1524, 2025.

Loong Kuan Lee, Thore Gerlach, Johannes Knaute, Florian Gerhardt, Patrick Völker, Tomislav Maras, Alexander Dotterweich, and Nico Piatkowski. Multi-objective quantum power system redispatch. In *IEEE DSAA*, pages 1–12, 2025.

Loong Kuan Lee, Geoffrey I. Webb, Daniel F. Schmidt, and Nico Piatkowski. Computing marginal and conditional divergences between decomposable models with applications in quantum computing and earth observation. *Knowledge and Information Systems*, 66(12):7527–7556, 2024.

Loong Kuan Lee, Geoffrey I. Webb, Daniel F. Schmidt, and Nico Piatkowski. Computing marginal and conditional divergences between decomposable models with applications. In *IEEE ICDM*, pages 239–248, 2023.

Geoffrey I. Webb, Loong Kuan Lee, Bart Goethals, and François Petitjean. Analyzing concept drift and shift from sample data. *Data Mining and Knowledge Discovery*, 32(5):1179–1199, 2018.